9

Disentangling

Credible estimates of all-else-equal relationships often represent the net effect of several mechanisms. This does not always provide enough information to answer important questions. We often want to know how each mechanism separately contributes to the overall effect. When that is so, we need to *disentangle*—breaking the estimated relationship down into its constituent parts.

For an example of why disentangling might be important, return to the debate about the role of Congressional parties we discussed in Chapters 6 and 8. Rohde (1991) suggested that parties exercised increasing influence over roll-call voting. Krehbiel (1993) countered that increased partisan sorting might instead explain increased party-unity voting. This dispute is normatively charged. Rohde's account raises concerns about a decline in democratic representation. Are members reducing the attention they pay to their constituents' preferences in order to do the party's bidding? Krehbiel's explanation does not suggest such a conclusion. How normatively troubling the over-time trends are depends, at least in part, on what caused them. How much of the increase in party-line voting is due to increased party discipline, and how much to increased ideological sorting? That question calls for disentangling quantitatively comparable estimates of the influence of these two mechanisms.

Disentangling involves distinguishing multiple mechanisms from one another. Assessing a distinguishing implication of a mechanism requires only one research design. But disentangling asks for more. We want to use implications of each mechanism to figure out how it is contributing to an empirical relationship. This requires multiple research designs, commensurable with multiple distinguishing implications.

To see the distinction, think back to the study by Ansolabehere, Snyder, and Stewart (2001) that we discussed in Chapter 8. They show evidence of

party influence by comparing members from different parties with the same NPAT scores. Those members vote differently, which they would not do were voting purely ideological. So this analysis distinguishes the party discipline mechanism. But it cannot speak to the relative importance of party influence and representative ideology. That would require disentangling these mechanisms, something their approach was not designed to do.

At first glance, disentangling seems like it might require the tools of causal mediation analysis (Imai et al., 2011). That might be worrying, since the identification assumptions for such an analysis are so strong (Green, Ha, and Bullock, 2010). But, while causal mediation analysis is one useful tool for disentangling, it is not the only one. Indeed, our examples in this chapter all involve disentangling through methods other than causal mediation analysis, though mediators do sometimes play a role.

9.1 Accountability and Competence in Elections

As we discussed in Chapter 8, an extensive literature uses term limits as a device to home in on the electoral incentives mechanism. Across a pair of papers studying the effect of term limits on incentives, Besley and Case (1995, 2003) uncover a puzzle. In their first paper, Besley and Case find evidence of an effect of term limits on electoral incentives. In a difference-in-differences analysis using data from the American states between 1950 and 1986, they show that per capita spending and taxes were higher under term-limited governors than non-term-limited governors. In their 2003 follow-up paper, Besley and Case repeat the analysis on a sample that extends through the mid-1990s and find that the effect of term limits has gradually disappeared over time. They conclude that "it seems likely that some omitted variable is responsible for the change in behavior observed for governors working under a term limit. This is an area ripe for future research" (Besley and Case, 2003, p. 55).

Alt, Bueno de Mesquita, and Rose (2011) argue for a resolution of the puzzle that combines a version of the model of electoral accountability we discussed in Chapter 7 with some institutional knowledge about the nature of gubernatorial term limits in the United States. Recall that the accountability model suggests two ways in which elections affect the quality of governance—not only the incentive mechanism that is Besley and Case's focus, but also an electoral selection mechanism. The key institutional fact is a change over time in the number of terms allowed. In the early postwar years, the majority of term-limited governors served under so-called "no succession" laws,

limiting them to a single term. By the mid-1980s, virtually every state with gubernatorial term limits had a two-term limit, with most of the changes coming in the late 1960s and later. Today, Virginia alone retains a one-term limit.

How does combining the two mechanisms with the institutional fact offer a potential resolution of the puzzle? Let's start by thinking about the identifying variation in Besley and Case's difference-in-differences studies. They regress their outcome variables against an indicator variable for whether the governor was term limited in a given year along with state and year fixed effects. So their estimates reflect how the outcome differed, on average, under a term-limited governor versus a non-term-limited governor within a state. To get a sense of what's going on, think about a state that switched from a one-term to a two-term limit (which is typical). In such a state, governors who aren't term limited are in their first term. Governors who are term limited are a mix of first-term governors (from before the policy change) and second-term governors (from after the policy change). So the difference-in-differences compares a weighted average of outcomes under first- and second-term governors who are term limited to the average outcome under first-term governors who are not term limited.[1]

In Besley and Case's first study, where the data go through only the mid-1980s, most term-limited governors are in their first term because states didn't change policy until late in the sample period. So that study is primarily comparing outcomes under first-term governors who are term limited (combined with a smattering of second-term governors who are term limited) to outcomes under first-term governors who are not term limited. These two groups of governors are differentiated by electoral incentives, but since both are in their first term, they are not differentiated by electoral selection. In Besley and Case's second study, which extends the data through the 1990s, many more term-limited governors are in their second term. So that study primarily compares outcomes under second-term governors who are term limited (along with some first-term governors who are term limited) to outcomes under first-term governors who are not term limited. Those governors are differentiated by both electoral incentives and electoral selection.

1. Things are actually somewhat more complicated than this because there are states that never switch, states that switch from no term limits to a two-term limit, and so on. But this discussion gives the general sense of what is going on.

This means that in their first study, Besley and Case's estimates primarily reflect electoral incentives. But in the second study, the estimates reflect two mechanisms pushing in opposite directions. Governors who are term limited have weaker incentives than governors who are not. This should make term-limited governors tend to perform worse. But governors who are term limited also happen to be in their second term and so have undergone electoral selection. This should make term-limited governors tend to perform better.

This situation is illustrated by the directed acyclic graph (DAG) in the top of Figure 9.1. Electoral incentives work through edge 2, while electoral selection works through edge 1, as a confounder. Moreover, those two mechanisms pull in opposite directions. So maybe the electoral incentive effect of term limits has not disappeared, but instead is being masked by electoral selection. To know if this is the case, we need to disentangle the two mechanisms, to see how much each is contributing to the estimated effect of term limits.

Alt, Bueno de Mesquita, and Rose disentangle by shutting down one mechanism at a time to see how the other operates. Their empirical strategy exploits variation in the length of gubernatorial term limits. To see the idea, suppose two otherwise identical states differ in how many terms a governor can serve. In state *A* governors can serve only one term, while in state *B* they can serve two terms. This sets up two different comparisons, one to identify each effect.

Compare first-term governors in each state. Each governor has been elected to office only once, so they have withstood the same level of electoral selection. However, the governor from state *A* cannot seek reelection, while the governor from state *B* can. Hence, the two governors differ in their electoral incentives, and comparing these two governors provides an estimate of the incentive effect, purged of electoral selection.

This comparison is illustrated in the bottom left DAG in Figure 9.1. The treatment of interest is whether or not a governor is term limited, since term limits affect incentives. The problem is that, when term limits apply at the second term, electoral selection says that politician quality is a confounder for identifying the incentive effect because higher quality politicians are more likely to make it to a second term. However, restricting attention to first-term governors, some of whom are term limited and some who are not, eliminates the confounding by eliminating the edge between term in office and term limits.

Next compare a first-term governor in state *A* to a second-term governor in state *B*. Both governors are term limited, so neither has electoral incentives.

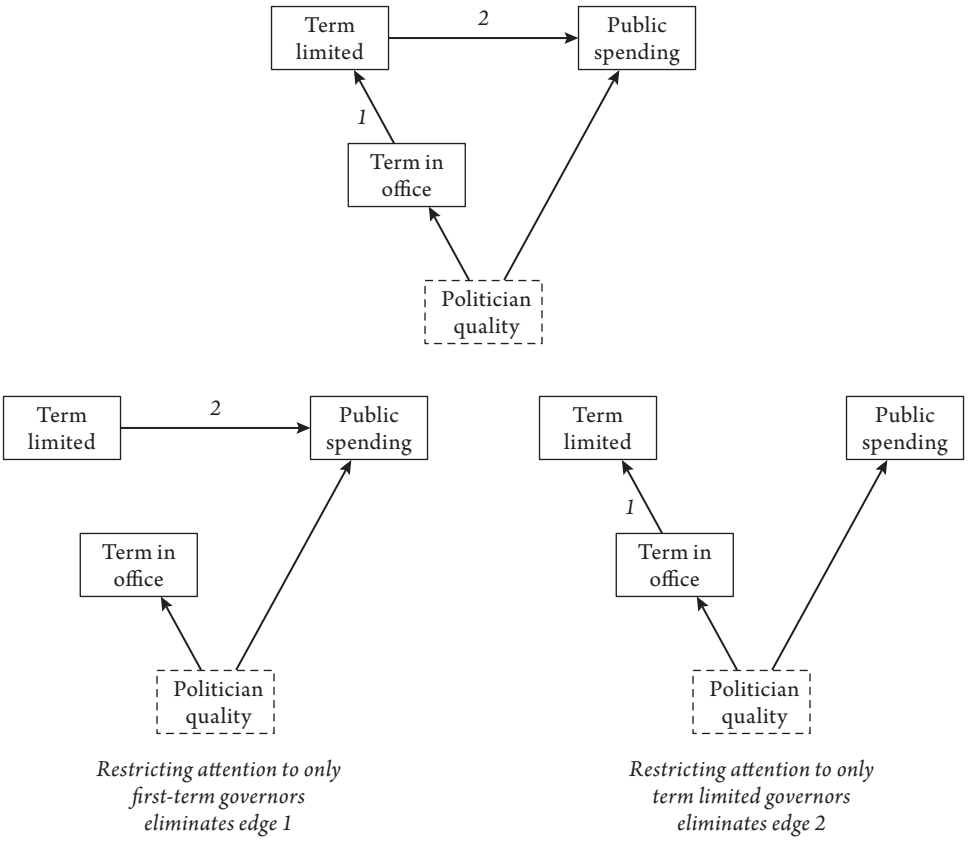

FIGURE 9.1. A DAG illustrating Alt, Bueno de Mesquita, and Rose's (2011) strategy for estimating the incentive effect by comparing first-term governors who are and are not term limited.

However, the second-term governor in state *B* has withstood a reelection challenge and thus has been subject to more electoral selection than the first-term governor in state *A*. Hence, comparing their performance provides an estimate of the electoral selection effect, purged of incentives.

This comparison is illustrated in the bottom right DAG of Figure 9.1. Here the idea is not to identify a treatment effect at all. Rather, the argument is that comparing governors who are all term limited breaks the causal link (running through the mediator of incentives) from term limits to performance. Hence, what is left is the bias due to electoral selection.

Alt, Bueno de Mesquita, and Rose make these comparisons within a difference-in-differences framework, while also controlling for a variety of time varying state characteristics. This means the variation exploited in estimation comes from states that change term limit laws. They drop the governor in

office at the time the term limit law was changed, to avoid the risk that the change in the law was related to that specific governor. In addition to studying the outcomes Besley and Case studied (i.e., per capita spending and per capita taxes), they also study borrowing costs for the state government and state economic growth (relative to national economic growth).

For all outcomes, Alt, Bueno de Mesquita, and Rose find both an electoral incentive effect and an electoral selection effect. Recall, to estimate incentives, they compare first-term eligible to first-term ineligible governors. Reelection eligible first-term governors have spending that is 6.5% lower, taxes that are 4% lower, borrowing costs that are 14 basis points lower, and economic growth that is .8 percentage points higher than term-limited first-term governors. To estimate electoral selection, they compare first-term term-limited to second-term term-limited governors. They find that second-term term-limited governors have spending that is 5% lower, taxes that are 3% lower, borrowing costs that are 14.5 basis points lower, and economic growth that is .5 percentage points higher than term-limited first-term governors. These empirical results suggest that both incentives and selection play a role in determining governance outcomes.

This disentangling indeed resolves the puzzle posed by Besley and Case. The effects of the electoral incentives and electoral selection mechanisms are roughly equal in magnitude, but pull in opposite directions. So when term limits switched from one to two terms, what appeared to be a vanishing effect of electoral incentives was actually the negative incentive effect of term limits being offset by the positive selection effect associated with being in the second term.

It is worth nothing that while Alt, Bueno de Mesquita, and Rose have an argument for having disentangled the effects of differences in incentives and differences in selection, there may still be more disentangling to do. For instance, the comparison of first-term term-limited to second-term term-limited governors estimates a quality difference that might reflect both electoral selection and learning-by-doing that occurs during a term in office. To disentangle these two mechanisms, future researchers would need some new research design.

9.2 Economic Shocks and Violence

Let's now take a more detailed look at one of the examples we discussed in Chapter 1, about how economic factors affect the likelihood of violent conflict. Early empirical studies on the topic used a just-controlling design

on a sample of country-years, regressing some measure of civil conflict (e.g., whether or not a civil war began, or the number of battle deaths) against a measure of the state of the economy (e.g., lagged GDP per capita, or unemployment), plus controls such as population, measures of rough terrain, whether the state was an oil exporter, measures of democracy, ethnic or religious fractionalization, natural resource endowments, and so on (see, for example, Fearon and Laitin, 2003; Collier and Hoeffler, 2004, and many others). These studies typically find a significant negative relationship between civil conflict and economic performance.

At least two issues complicate such studies. Most straightforwardly, the state of the economy cannot be treated as exogenous to civil conflict. This is especially true because of the clear reverse causality problem—the anticipation of civil conflict will affect the economy through (among many other pathways) investment and migration decisions.

Scholars have employed a variety of creative research designs in an attempt to more credibly estimate the all-else-equal relationship between economic factors and conflict. An early example was Miguel, Satyanath, and Sergenti's (2004) use of rainfall shocks in sub-Saharan Africa as an instrument for income. But probably the most prominent research design in this literature is a difference-in-differences design exploiting shocks to the value of a country's commodities.

Here's the basic idea. Start by measuring a fairly deep lag of each country's commodity exports. Take this bundle of commodities to represent the underlying natural commodity endowment of the country. For each commodity, calculate the world price each year. For most commodities, most countries are price takers. Hence, the interaction of the underlying commodity endowment of a country and the world prices of the commodities provides a year-to-year measure of exogenous economic shocks. This interaction can thus be used as a source of variation to credibly estimate the relationship of interest. Using this approach, Bazzi and Blattman (2014) find little evidence of a systematic relationship between economic shocks and civil conflict.

This takes us to the second complication, one that is particularly important for highly aggregated studies of the economy and civil conflict. Economic factors might affect conflict via many mechanisms (Fearon and Laitin, 2003; Humphreys, 2005; Bazzi and Blattman, 2014). So even a credibly identified all-else-equal relationship is difficult to definitively interpret. This is especially so since not all of the mechanisms linking economic factors to conflict pull in the same direction.

A model will help think through these complications. A mass of citizens, with size N, is deciding whether to join a rebel organization that is fighting for control of some territory or resource. To keep things simple, suppose that each citizen can create a fixed amount of violence, v. So if the proportion of citizens mobilizing is η, the amount of violence is $\eta \cdot N \cdot v$. Moreover, assume that the rebels win with probability η when η percent of citizens mobilize. This represents the idea that the more people who rise up against a government, the more likely that government is to fall.

The total spoils are B, evenly divided among the participants in the event of success. If individual i joins the rebels, her expected payoff is $p \cdot B_i$, where p is the probability the rebel group wins and B_i is her share of the spoils of victory. So if η percent of the people participate and the rebels win, each participant gets a benefit of $B_i = \frac{B}{\eta N}$.

If, instead, citizen i does not join the rebel group, she makes a payoff equal to her wage, w_i. The distribution of wages in the society is F.

B represents the economic and political returns from participating in a winning rebellion. So changes in B represent economic shocks that change the value of the economic spoils of conquest. And w_i represents a citizen's wages in the regular economy. Thus, changes to the distribution of the w_is represent economic shocks that affect wages.

Suppose citizen i believes a fraction η of other citizens will participate in the rebellion. Then citizen i's best response is to participate if

$$\eta \cdot \frac{B}{\eta N} > w_i,$$

or

$$w_i < \frac{B}{N}.$$

Because of the linearity assumptions in this simple model, a citizen's participation decision turns out to be independent of her belief about others' participation. This is because others' participation decisions enter both the probability of winning and the share of the rewards in ways that cancel with one another. (This is obviously not a robust feature of the model, but it is also not essential for the key results and is not meant to be representational.) Hence, the share of the population that participates in rebellion is the fraction of citizens with wage less than this participation threshold:

$$F\left(\frac{B}{N}\right).$$

The amount of violence is proportional to this quantity.

Two implications follow. First, suppose a shock increases the economic value of the prize, *B*. More people participate and there is more violence. Second, suppose a shock improves the distribution of wages, *F*, so that fewer people have wages below any fixed cutoff.[2] Then fewer people participate and there is less violence.

This model embodies two mechanisms by which economic shocks might affect violence. On the one hand, a weaker economy might diminish conflict by reducing the value of territorial conquest, weakening incentives for predation. On the other hand, a weaker economy might increase conflict by reducing wages, thereby reducing the opportunity costs of mobilization. So even a credible estimate of the all-else-equal relationship between the state of the economy and civil conflict might be estimating a mixture of these competing effects.

As Bazzi and Blattman point out, these mechanisms pull in opposite directions, so there are at least two ways one might interpret the null result. One possibility is that the putative mechanisms linking economic factors to civil conflict are unimportant. Maybe there really is nothing to see here. But another possibility is that multiple mechanisms are doing important work, with their offsetting effects resulting in a null net effect. To assess which of these accounts is more plausible, we need to disentangle the mechanisms.

Dal Bó and Dal Bó (2011) study a theoretical model that helps structure such a disentangling. They augment a canonical model of trade to include predatory conflict. The model differentiates between two different types of commodities: *labor intensive* and *capital intensive.* Negative shocks to the price of labor-intensive goods have a big effect on wages—that is, they are similar to a worsening of the distribution of wages, *F*. Consequently, such shocks primarily activate the opportunity costs mechanism and thus lead to an increase in conflict. Negative shocks to the price of capital-intensive goods have little effect on wages, but do affect the value of the prize over which people fight—that is, they are similar to a decrease in the value of the resource being fought over, *B*. Consequently, such shocks primarily activate the predation mechanism and thus decrease conflict.

Since shocks to labor- and capital-intensive goods primarily activate different mechanisms, they should affect conflict in opposite directions. This is an instance of the worry we noted earlier: if a country's economy is made up

2. Formally, consider an improvement in the sense of first-order stochastic dominance.

of a mix of capital- and labor-intensive goods, then finding that the average effect of a commodity price shock is zero might reflect these offsetting effects, rather than reflecting that the economy doesn't matter for conflict outcomes. But this way of stating the worry also suggests a path forward: disaggregate and look at labor-intensive and capital-intensive goods separately.

Dube and Vargas (2013) use this insight to take a step in the direction of disentangling, using data on the conflict in Colombia. They start by arguing that two of Colombia's primary exports—oil and coffee—fit Dal Bó and Dal Bó's distinction between capital- and labor-intensive goods, respectively. They then follow the approach described earlier for cross-country studies exploiting commodity price shocks, but at the subnational level. That is, they measure the coffee and oil intensivity of each municipality in Colombia. They interact these fixed municipal endowments with the world prices of coffee and oil. They also have municipality-level data on conflict events. They can now study the effect of commodity price shocks in a difference-in-differences framework. That is, they ask how changes to world coffee prices affect conflict outcomes in coffee-intensive municipalities relative to non-coffee-intensive municipalities, and likewise for oil. The argument is that the differential response of violence to coffee price shocks in coffee-intensive versus non-coffee-intensive municipalities provides an estimate that primarily (though perhaps not exclusively) reflects the opportunity costs mechanism. Similarly, the differential response of violence to oil price shocks in oil-intensive versus non-oil-intensive municipalities provides an estimate that primarily reflects the predation mechanism.

This idea is illustrated in Figure 9.2. The left-hand side of the figure illustrates the argument that coffee price shocks affect violence through the opportunity-cost mechanism (i.e., by changing wages) but not through the predation mechanism, while oil price shocks do the opposite.

The evidence supports both mechanisms. When world coffee prices rise, violent events decrease in coffee intensive municipalities relative to non-coffee-intensive municipalities. And when world oil prices rise, violent events increase in oil-intensive municipalities relative to non-oil-intensive municipalities.

Dube and Vargas also elaborate, looking for further ways to assess whether their research designs are in fact identifying the effects of the different economic mechanisms. For instance, they look at a key mediator, showing that shocks to coffee prices are associated with changes in wages in coffee intensive-municipalities relative to non-coffee-intensive municipalities, while

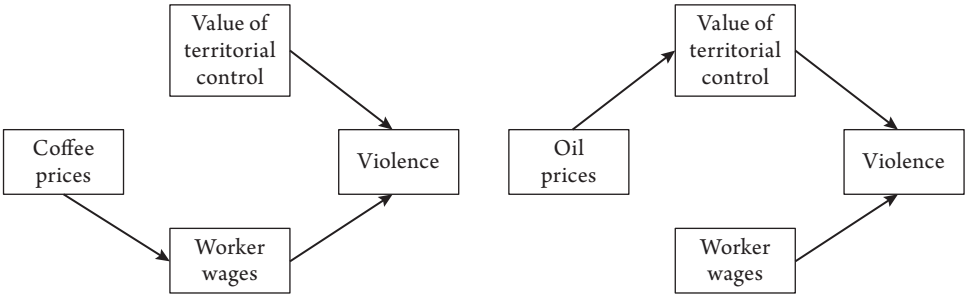

FIGURE 9.2. DAGs illustrating Dube and Vargas's (2013) strategy for disentangling the opportunity cost and predation mechanisms.

shocks to oil prices are not associated with changes in wages in oil intensive municipalities relative to non-oil intensive municipalities. Hence, the data from Colombia agree with Dal Bó and Dal Bó's underlying model of the economy, as well as with the interpretation that economic shocks matter for conflict. Moreover, the results reveal the off-setting effects of the two mechanisms, providing a possible explanation for why those effects can't be seen in regressions that pool together the effects of shocks to many different commodities.

It is worth noting that Dube and Vargas's strategy may not provide a complete disentangling. Figure 9.2 presents an idealized version of the world. But shocks to coffee prices, while working primarily through opportunity costs, may have at least some effect on the value of predation. And shocks to oil prices, while working primarily through predation, may have at least some effect on opportunity costs. So, while these research designs are informative about the competing effects, we can't, in any simple way, compare the magnitudes of the estimates they provide to ask whether the two mechanisms are exactly offsetting.

A possible concern about Dube and Vargas's study as a general explanation of Bazzi and Blattman's null result is that its geographic scope is limited to Colombia. So it is important to note that a subsequent literature has combined Dube and Vargas's empirical strategies for disentangling with newly available data. These studies expand the geographic scope, without giving up commensurability with the distinct implications of the opportunity-cost and predation mechanisms. For instance, Berman et al. (2017) combine fine-grained data at the 50 km by 50 km grid-cell level on violence and

mining intensity for fourteen different minerals across all of Africa with data on mineral price shocks to show evidence consistent with the predation mechanism. And Berman and Couttenier (2015), again at the grid-cell level, use measures of agricultural production to show that commodity price shocks lead to increased conflict in sub-Saharan Africa in ways consistent with the opportunity-cost mechanism. So this literature also illustrates how well-identified studies, combined with both theory and the availability of new data, can help spur the ideas and machinery needed to broaden the settings to which accumulated knowledge can be applied.

9.3 Sources of Incumbent Success

In an agenda-setting paper, Erikson (1971) pointed out that incumbents win reelection at strikingly high rates. This observation raised a normative concern: Do high reelection rates indicate that incumbents use the perquisites of office—for example, greater access to campaign resources, gerrymandering, or public attention—to insulate themselves from electoral threat? If so, elections may be ineffective at generating good governance outcomes. Cox and Katz (2002, p. 7) nicely summarize the worry:

> Whenever the resources of public office are used to insulate individual politicians from electoral risk, their accountability to their constituents is weakened. . . . Thus, insulation from electoral risk of the kind suspected would, at a single stroke, debilitate the two fundamental accountability relationships of a democratic system of government.

This normative concern has attracted the attention of journalists, pundits, and activists, and has sparked an enormous empirical literature attempting to estimate the so-called incumbency advantage.

The literature on the incumbency advantage discusses several mechanisms that might underlie incumbent success. First, districts have partisan leanings and incumbents are often from the party the district favors. This *party match* mechanism naturally gives rise to high reelection rates for incumbents (Gelman and King, 1990). Second, high incumbent reelection rates may reflect *electoral selection*. Incumbents become incumbents by appealing to voters. Thus, the average incumbent may have characteristics that appeal to voters more than those of the average challenger (Samuelson,

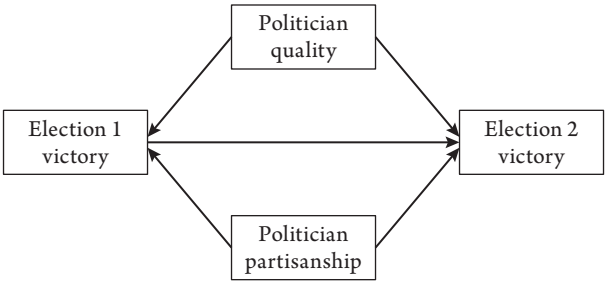

FIGURE 9.3. A DAG illustrating three sources of the incumbency electoral success.

1984; Zaller, 1998; Ashworth and Bueno de Mesquita, 2008). Third, incumbents may be *insulated* from electoral challenges through the perquisites of office.[3]

This insulation mechanism motivates the normative concerns we discussed earlier. The first two mechanisms do not have the same kind of normative implications—indeed, it might be normatively appealing if incumbents win reelection at high rates because voters tend to like them, whether because of partisan affinity or positively valued characteristics.

We illustrate the basic ideas in the theoretical literature with the DAG in Figure 9.3. The treatment is winning a first election (i.e,. becoming an incumbent) and the outcome is vote share in the next election. The direct effect runs through the insulation mechanism. The other two mechanisms appear as confounders. The partisanship of a politician affects both whether they win a first election and how they perform in the second election. This is the party match mechanism. Similarly, the quality of a politician affects whether they win a first election and how they perform in the second election. This is the electoral selection mechanism.

Fowler (2016) develops an empirical strategy to disentangle these sources of incumbent electoral success. Specifically, he estimates the extent to which incumbent electoral success can be attributed to three mutually exclusive factors: party match; candidate quality, which is defined as those fixed attributes of candidates that make them more appealing to voters than nonincumbents

3. The literature also highlights the possibility that incumbents scare off high-quality challengers (Cox and Katz, 1996). Such scare-off could arise as a consequence of any of the three mechanisms already discussed and so are not separately featured in our discussion.

of the same party (capturing electoral selection); and the electoral benefits obtained by holding office (capturing insulation).[4]

Fowler uses a multistep empirical analysis to disentangle these three components of the incumbency advantage. He starts by regressing Democratic vote share on an incumbency variable, coded to take the value of 1 if a Democratic incumbent seeks reelection, 0 if no incumbent runs, and −1 if a Republican incumbent seeks reelection. The regression also includes year fixed effects to account for national trends in partisanship. The estimated coefficient on the incumbency variable, β_1, is the average additional vote share that incumbents receive relative to nonincumbents from the same political party. This difference is an omnibus measure of incumbents' electoral advantage, capturing the sum all three factors just defined.

We have emphasized that disentangling requires multiple research designs. In this case, the three effects sum to β_1, by construction. Thus, Fowler needs two research designs above and beyond his estimate of β_1—having estimated any two of the component effects, the third can be deduced by subtracting the other two from β_1. Note that this last step in the decomposition is possible only because of the assumption that no other mechanism is at work. We return to this point.

Fowler's first research design augments the cross-sectional regression with state-by-decade fixed effects, to account for the district's normal vote. That is, the state-decade fixed effects capture the average performance of Democratic candidates in the state during a particular decade. This strategy removes the party match component of incumbent success and recovers a parameter, β_2, that is the sum of electoral selection and insulation.[5] Subtracting β_2 from β_1 thus reveals the size of the party match component.

Fowler's second research design uses a fuzzy regression discontinuity. This aims to estimate insulation alone, purged of electoral selection and party

4. Fowler refers to these three mechanisms as party match, characteristic selection, and officeholder benefits, respectively.

5. Gelman and King (1990) estimate the incumbency advantage purged of district partisan leanings by regressing candidate vote share on a dummy for incumbency and a measure of the normal vote—that is, the expected vote of a candidate's party in a district. This strategy estimates the extent to which an incumbent candidate from a given party in a given district performs better than would be expected for a candidate from the same party in an open-seat election in the same district. Such an estimate includes both the insulation and electoral selection mechanisms. Fowler's β_2 is roughly comparable to the estimate from Gelman and King's regression, though it makes use of fixed effects rather than a direct measure of the normal vote.

match. Following Lee (2008), he uses Democratic share of the two-party vote as the running variable. There is a discontinuity in the likelihood of having a Democratic incumbent at 50%. Fowler uses a close election outcome in the previous election as an instrument for incumbency in the next election. (The design is fuzzy because some incumbents do not run for reelection.)

Fowler's regression discontinuity delivers a coefficient, β_3, which is the difference in next-election Democratic vote share between races with barely elected Democratic incumbents and races with barely elected Republicans incumbents. Recall that the incumbency variable is 1 for Democratic incumbents and -1 for Republican incumbents, so the difference between Democratic and Republican incumbents is 2. Thus Fowler interprets the estimate as twice the insulation component of the incumbency advantage, purged of party match and electoral selection.

There is an important caveat to this RD analysis regarding commensurability. Continuity of potential outcomes at the election threshold implies that, in the limit, winners and losers of close elections do not differ systematically *from one another* in terms of individual quality or party match. However, for the RD approach to identify insulation purged of electoral selection and party match, it must be that winners of close elections do not differ systematically in terms of individual characteristics or party match *from those they could subsequently face as challengers.*[6] As we discuss in Chapter 10, this condition is not an implication of the standard continuity assumption for RD but requires additional assumptions. For now, we simply take such additional assumptions as granted, returning to them in the next chapter.

With the estimates of β_1, β_2, and β_3 in hand, Fowler can disentangle the incumbency advantage into its three components as follows:

$$\begin{aligned} \text{Party match} &= \beta_1 - \beta_2 \\ \text{Electoral selection} &= \beta_2 - \beta_3 \\ \text{Insulation} &= \beta_3. \end{aligned} \tag{9.1}$$

Fowler's estimates suggest that all three factors play an important role in explaining incumbent success in elections. His primary analysis is for US

6. There is a subtlety here—insulation includes scare-off effects. As such, it is important to distinguish between the pool of politicians who could run as challengers from those who actually choose to do so.

Senate elections between 1926 and 2010, though he also extends the analysis to other offices covering more than 90,000 elections. In the Senate, the total incumbency effect is estimated at 12.3 percentage points, of which party match accounts for 5.3 percentage points, electoral selection accounts for 3.3 percentage points, and insulation accounts for 3.7 percentage points. Looking beyond the Senate to seven different electoral settings, he finds that the share of incumbent success due to each component varies but that party match is typically the largest factor, with the lone exception of gubernatorial elections.

With this disentangling, Fowler can speak to the larger normative debate that motivates much of the literature. Notably, according to Fowler's analysis, insulation explains less than half of incumbent success, suggesting that the presence of an incumbency advantage is not as damning of the US political system as previously thought.

As we've discussed, the regression discontinuity design Fowler uses requires additional assumptions to purge electoral selection. If electoral selection is not purged from β_3, then not only is the estimate of insulation called into question, but so is the estimate of electoral selection, threatening the overall validity of the disentangling. Even if such concerns about β_3 are valid, however, Fowler has still disentangled the party match component, which he shows to be the most important factor in the overall incumbency advantage.

There is one more potential issue with Fowler's strategy for disentangling electoral selection and insulation. His approach assumes that those mechanisms operate in an additively separable way. But there could be an interaction between incumbent quality and the perquisites of office. To be concrete, let θ be a measure of the incumbent's quality, and suppose that the increase in vote share due to officeholder benefits is equal to $\gamma\theta$. Substantively, this would represent a situation in which higher quality incumbents can more effectively take advantage of the various perquisites of office.

If there is such an interaction, the average increase in vote share due to officeholder benefits in the population of incumbents is $\gamma\mathbb{E}(\theta \mid \text{incumbent})$. But this is not what the RD estimates. The RD analysis only includes incumbents who barely won office, a group likely to have a different distribution of quality than the full population of incumbents. Hence, the RD estimates a LATE, namely $\beta_3 = \gamma\mathbb{E}(\theta \mid \text{close election})$. Since electoral selection is estimated as the difference $\beta_2 - \beta_3$, the difference between the LATE and the population average of the officeholder benefit affects the estimate of the effect of both mechanisms. In particular, the estimate of electoral selection will also include the quantity $\gamma[\mathbb{E}(\theta \mid \text{incumbent}) - \mathbb{E}(\theta \mid \text{close election}]$.

This potential difference between the RD estimand and the average effect of insulation reflects a broader issue. The three analyses in Fowler's disentangling are based on coefficients identified off of different subpopulations of districts: β_1 is identified off of all races, β_2 is identified off of offices that changed hands within state-decades, and β_3 is identified off of races that follow close elections. Even if there are no interactions across the mechanisms, these different subpopulations may not be comparable. Fowler addresses this concern with an analysis that reweights the fixed-effect and RD portions, to correct for the differences in subpopulations. His results are not sensitive to this reweighting.

9.4 Gleanings

Disentangling makes sense when two things are true. First, you have good reason to believe that an existing estimate reflects the combined impact of multiple mechanisms. Second, understanding the relative contribution of those mechanisms is of interest. When both of these conditions are met, you face a challenge. Disentangling involves distinguishing multiple times. So it is feasible only when you have multiple research designs.

The papers we've discussed exemplify two different motivations for understanding the relative contribution of different mechanisms.

The studies by Alt, Bueno de Mesquita, and Rose and by Dube and Vargas both start with a surprising existing finding. While it was possible no mechanism of interest was at work, it was also possible that two mechanisms had offsetting effects. For Dube and Vargas, the puzzling fact was the absence of a relationship between commodity price shocks and conflict. This left open the possibility that economic factors were unimportant for conflict outcomes. By studying the effects of shocks to labor-intensive and capital-intensive commodities, Dube and Vargas showed that it is likely that those null results were a consequence of two substantively important but offsetting economic mechanisms—opportunity costs and predation. Both of these mechanisms are reflected in a measure of shocks to the aggregate value of a country's commodity bundle. Something similar is going on in Alt, Bueno de Mesquita, and Rose's effort to understand Besley and Case's surprising null result.

Fowler's motivation was normative. Some of the mechanisms that might lead incumbents to win at high rates are normatively troubling. Others are potentially desirable. Knowing how much of the incumbency advantage is due to each is thus important for our overall take on democratic performance.

These papers also differ in the extent to which they fully disentangle the mechanisms. Alt, Bueno de Mesquita, and Rose disentangle mechanisms using the same data that generated the aggregate findings. As such, it makes sense to add up their component estimates and compare them to the aggregate estimates. Fowler's estimates have an even more direct relationship with aggregate estimates, since replicating them is the first step in his disentangling. The situation for Dube and Vargas is different. The cross-country results they are responding to are about shocks to countries' aggregate commodity bundles. There is no reason to think that Dube and Vargas's estimates of the effect of coffee shocks and oil shocks on conflict in Colombia will add up to any quantity directly derived from the estimated effects of aggregate commodity shocks across countries. In this sense, they have not literally disentangled those prior results. Nonetheless, their exercise is in the spirit of disentangling. We can make better sense of the existing aggregate null findings once we see evidence that the opportunity cost and predation mechanisms, both of which can be activated by commodity shocks, pull in opposite directions.

Finally, it is worth noting an interesting feature that differentiates Fowler's analysis from the other two. Fowler estimates three quantities with three different research designs. But those research designs are not independent of one another. The research designs for party match and electoral selection both involve the quantity β_2, and the research designs for electoral selection and insulation both involve the quantity β_3. There is a trade-off here. By cleverly combining estimates, Fowler is able to maximally exploit the data to disentangle several mechanisms. But this comes with a risk. Any problem in estimating either one of β_2 or β_3 compromises the credibility of his method to disentangle two of the mechanisms of interest.